

Controlling information systems development: a new typology for an evolving field

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Abstract. *This study explores how and why information systems development (ISD) approaches differ in their tactics to control the behaviour of development teams. Drawing from prior research on organisational control, we conduct qualitative case studies at four organisations to examine the extent that traditional and contemporary systems development approaches can highlight a unique combination of control attributes. Based on the data collected, we build an ISD control typology that differentiates ISD approaches on the basis of control objectives (product or process focused) and control practices (preventive or detective/corrective focused). The proposed typology can be used by practitioners to inform and guide more effective systems development control choices while providing researchers with a new model that recognises the unique control aspects of today's systems development approaches.*

Keywords: information systems development, control, agile, waterfall, development approach

INTRODUCTION

Effective control over information systems development (ISD) is widely viewed as an important contributor to achieving project and organisational objectives (Henderson & Lee, 1992; Sia & Neo, 1997). A key determinant of effectively managing ISD projects stems from the development approach, which is viewed as 'a cornerstone of disciplinary practice, a means of controlling processes and results' (Orlikowski, 1991, p. 16). Though prior research has highlighted a need for advancement in control conceptualisations in order to extend knowledge in the field (Kirsch, 2004), the role of control in systems development approaches remains ambiguous, particularly in light of recent ISD trends such as agile.

Organisational control has been widely studied in the management literature and broadly refers to a process in which a person or group intentionally affects the behaviour of another person or group, with the objective of achieving goals (Davis, 1940; Tannenbaum,

1962; Flamholtz *et al.*, 1985). In the context of information systems (IS), control research has predominantly focused on how different categories of control are determined by the project context and organisational circumstances (Kirsch, 1996; Choudhury & Sabherwal, 2003). However, much of the IS control literature has yet to consider the unique behavioural and managerial aspects of today's emerging systems development initiatives. For example, there is a perceived trade-off between control and flexibility as a central dynamic in contemporary development trends. Some argue that the fundamental design views of emerging approaches contradict control (Augustine *et al.*, 2005), while others find that less formal control approaches are not a significant predictor of systems development success (Misra *et al.*, 2009). As a consequence, managers are uncertain about the control implications associated with using different ISD approaches. There is ambiguity regarding the control differences associated with various approaches and confusion about the aspects of control that might lead a manager to choose one development approach over another (Fitzgerald, 1996; Augustine *et al.*, 2005; Nerur *et al.*, 2005).

This study aims to answer the research question: *how and why does control differ between ISD approaches?* The objective of this paper is to develop a theoretically and empirically derived typology of ISD control that articulates the fundamental control dimensions that differentiate systems development approaches. Practitioners will benefit from this research by deriving a clearer view of how control is embedded in the development process; this clarity may enable organisations to more effectively manage ISD initiatives by using control more appropriately. From a research perspective, an improved understanding of the relationship between control and systems development is important to enable more sophisticated theorising of the organisational implications of ISD activities.

The remainder of this paper is structured as follows. The next section will provide an overview of the ISD control literature. This will be followed by an overview of the research methodology and a presentation of the study's results. Next, the typology of ISD control will be proposed. Finally, the paper will conclude with a discussion of the study's implications, limitations and opportunities for future research.

CONTROLLING SYSTEMS DEVELOPMENT

Systems development control refers to the process used to influence project stakeholders (e.g. developers, users and managers) to behave in a way that is consistent with the achievement of project and organisational goals. Past research on systems development control largely focuses on the organisational, project and individual characteristics that influence the types of controls that are chosen to be used on ISD projects. For example, where behaviour observability and outcome measurability of ISD project tasks are high, past research has suggested that controls consisting of managers mandating the output levels of subordinates are more prominently used (Kirsch, 1996). Research in this area also focuses heavily on the performance outcomes of ISD control. Desirable outcomes such as quality (Maruping *et al.*, 2009), speed (Jiang *et al.*, 2004) and creativity (Fitzgerald, 1996) are examined in relation to the employment of particular categories of ISD control.

The control framework used most prominently in systems development research is control modes, consisting of behavioural, outcome, clan and self control (Ouchi, 1978; 1979; Kirsch, 1996; 1997). Each type of control represents a distinct approach to influencing the activities of organisational actors: behavioural control via management monitoring, outcome control via defined output requirements, clan control via social norms within teams and self-control via individual empowerment. Related research finds that organisations operating in dynamic, changing environments will modify the collection of controls utilised throughout the lifecycle of an IS project as priorities change; the result is an evolving portfolio of control mechanisms for IS projects (Kirsch, 1997; Choudhury & Sabherwal, 2003). The control mechanisms themselves can take a variety of forms, including formal policies and procedures, training programs and technical tools.

However, few studies of IS control conduct detailed investigations of the unique characteristics of particular control mechanisms. For example, one widely used method to control the ISD process is the establishment and utilisation of a development methodology and development approach. Consistent with livari *et al.* (2000), we define development methodologies as the methods and concepts that guide the building of a system, while development approaches represent the higher level goals and principles that form a class of methodologies sharing common features. For example, under this view, we consider agile development to be a development approach, with eXtreme Programming and Scrum representing two specific development methodologies that share common agile characteristics. Development approaches establish an important foundation from which the development activities are initiated, including philosophies on project team interactions, the role of the customer in project activities and development priorities. In this study, we choose to focus primarily on development approaches as a means to compare the high-level goals and principles of systems development as they relate to the concept of control, rather than the specific development methods.

In the relatively limited research that has considered the control aspects of ISD approaches, differing perspectives are presented. Westrup (1993) argues that management control can be exerted via development approaches by demanding specified outcomes at particular project stages through development tools such as computer-aided software engineering (CASE) and extensive staff training. Orlikowski (1991) argues that development approaches and methods represent a form of cultural control over developers as they force an 'internalization of . . . assumptions, values, and interests' (p. 17). Several other researchers have extended the use of control theory into software development environments (Henderson & Lee, 1992; Choudhury & Sabherwal, 2003; Nidumolu & Subramani, 2003), and more recently researchers have begun examining control issues in flexible software development (Harris *et al.*, 2009; Maruping *et al.*, 2009). However, a paucity of research still exists in regard to the underlying control aspects that differentiate ISD approaches.

In general, development approaches are perceived to exhibit a range of unique characteristics (Baskerville *et al.*, 2007; Kautz *et al.*, 2007; livari & livari, 2011); however, as the focus of this study is the investigation of control phenomenon in practice, we concentrate on approaches that are perceived to differ on that basis. As such, agile and waterfall approaches are selected on the grounds that they are both widely used in practice (Nerur *et al.*, 2005) and

are perceived to be conflicting in their core strategy and philosophy, particularly relating to their approach on control: waterfall is commonly perceived to be structured and controlled, while agile is viewed as being flexible and uncontrolled (Augustine *et al.*, 2005; Coram & Bohner, 2005). Where agile is viewed as an example of the evolving and emerging face of the systems development field, waterfall is more commonly seen as the staid and traditional approach of the past. By examining control in both agile and waterfall, this study can consider the past, present and future of ISD control.

Traditional software development approaches, such as waterfall, are highly structured and sequential in nature; the early dominance of these methods in practice is attributed to advocates that argued for development to be completed via discrete phases, whereby requirements analysis, design, testing and implementation are individually completed in a pre-defined sequence (Royce, 1970; Boehm, 1976). Using these plan-driven, structured methods, customers are primarily involved in the requirements gathering but have reduced participation in subsequent phases. In recent years, a movement to encourage more flexibility to respond to changing customer needs in software projects has resulted in a number of more flexible and agile development approaches including methodologies such as eXtreme Programming, Scrum, Dynamic Systems Development Method and Feature-Driven Development. Agile development is based around the concept of collaborative, self-organising teams that iteratively deliver working software (Martin, 1991; Beck *et al.*, 2001). Agile projects take a minimalist approach to design and documentation, relying instead on adaptability, flexibility and responsiveness (Fitzgerald *et al.*, 2006; Lee & Xia, 2010). Project team members tend to have only loosely defined responsibilities, and customers are encouraged to be actively involved in the development process (Nerur *et al.*, 2005). Agile development methods allow software teams to increase flexibility by allowing them to embrace and respond to changing environments (Beck & Andres, 2004).

Although control is perceived to differ between agile and waterfall approaches, this view is largely anecdotal and ambiguity exists as to the specific dimensions where control is unique. By conducting an empirical examination of agile and waterfall projects, we sought to consider the specific aspects of control that could contribute to an enhanced understanding of differences between ISD approaches.

RESEARCH METHODOLOGY

In order to provide useful insights related to control phenomena, researchers have called for an increased focus on fieldwork and observation of organisational practices (Packendorff, 1995). In response, we conduct a set of qualitative case studies to explore why and how ISD approaches differ on the basis of control phenomena.

Study design

Since organisations use a range of controls and developers may have experience with different ISD approaches, a multiple case study design was considered the most effective research

design in order to compare control phenomena. The unit of analysis for this study is at the development project level. Though control phenomena exist at the organisational, project and individual level (Ouchi, 1978; Flamholtz *et al.*, 1985), our focus is on examining how control varies across projects that use different development approaches. By concentrating on the project level, it allows this study to more closely examine the execution-oriented control principles that are utilised in practice, thus resulting in a more reliable and valid typology of systems development control.

Four organisations were selected to participate in this research. The first organisation, referred to as UniCan, was selected as a pilot study, primarily on the basis of access and geographical proximity. Pilot studies are occasionally used in case study research as a means to refine the data collection process, clarify the study context and streamline field inquiry logistics (Dube & Pare, 2003; Yin, 2009). The data collected from UniCan was used for these purposes, but the data itself was excluded from the data analysis process because of subsequent adjustments made to the data collection approach. UniCan is a mid-sized Canadian University with a range of undergraduate, graduate and doctoral programs. Interviewees were involved in the systems development activities at the University, working on a variety of web and application development activities related to areas such as student registration and awards. UniCan utilises a hybrid systems development methodology, including techniques characterising both agile and waterfall methodologies.

The three remaining organisations were selected on the basis of fundamental differences in their approach to systems development. LargeMan is a multinational manufacturing conglomerate based in the USA. The company has offices in 60 countries around the world and over 75 000 employees. The location used for this study is in Ontario, Canada, where the local development team is responsible for the development of an inventory tracking application for the organisation's products. LargeMan uses a waterfall systems development approach, characterised by structured, sequential development cycles, detailed project budgeting and tracking, and formal approval of phase deliverables. Next, InfoOrg is a large information and technology company with approximately 50 000 employees. The location used for this study is in British Columbia, Canada, where a customer-facing finance application is developed and supported by a local project team. Although the core application utilises a primarily waterfall approach, smaller supplementary development projects have adopted an agile approach, integrating daily 'stand-up' meetings, frequent delivery of working code and short, iterative development cycles. The final case study organisation is a software development firm referred to as OntCode that develops software for financial services, retail and manufacturing customers in Ontario. OntCode uses an agile development approach, characterised by iterative development cycles, collective code ownership and pair programming techniques.

In order to provide a preliminary framework to guide the data collection activities, a thorough review of organisational and IS control literature, including both conceptual and empirical work, was conducted to distinguish fundamental aspects of the control process. Twelve distinct dimensions of control were identified. Each dimension represents a higher level categorisation of specific control mechanisms that share a common set of characteristics. For example, the

Table 1. Control categories, dimensions and definitions

Control category	Control dimension	Definition
Control objective	Speed	The minimisation of time to complete a systems development project.
	Budget	The effective planning and monitoring of the financial costs associated with a development project.
	Functionality	The delivery of system functions that best meet the requirements of customers.
	Regulatory compliance	The adherence of the project to all applicable internal and external laws and regulations.
	Product quality	The creation of a system that maximises reliability, accuracy and completeness of operation.
Control practice	Risk reduction	The reduction of risk associated with executing the development project.
	Active monitoring	The periodic inspections and confirmations conducted by project stakeholders to verify the execution and progress of development activities.
	Enforcement	The disciplinary repercussions faced by project participants who contravene defined development project guidelines set by management.
	Standardisation	The definition of a series of established and accepted development activities that are conducted on each project.
	Personnel interactions	The ongoing exchange of project information, activities and development status among project stakeholders.
	Remediation	The identification and rectification of issues preventing the effective and efficient execution of systems development project activities.
	Prioritisation*	The establishment of a defined precedence of outstanding development project activities.

*Prioritisation was identified during fieldwork as an emergent control practice but is presented here for completeness of the control categories referred to in the study.

'standardization' dimension could include a variety of control mechanisms ranging from documented security standards to an informal project management approach. In an effort to categorise these dimensions into higher level categories, we asked nine IS professionals (five academics and four practitioners) with extensive systems development experience to categorise the dimensions as either control objectives or control practices. Moore & Benbasat's (1991) guidelines for confirmatory card sorting were used to structure this exercise. The results of the card sort identified the following items corresponding to one of two distinct categories of control (see Table 1).

Data collection

Interviews were conducted at the case sites during June and July 2009. A total of 26 semi-structured interviews were completed with participants in a range of systems development roles. A similar number of interviewees were selected from organisations representing each development approach: seven from the waterfall organisation (LargeMan), seven from the agile organisation (OntCode) and eight from the hybrid organisation (InfoOrg). An additional four participants were interviewed from UniCan for the pilot study. The nature of the

interview data collection was both a retrospective account of participant experiences as well as perspectives related to ongoing projects. A formal interview protocol was utilised based on the preliminary framework, but the interviewers remained open to probing into participant responses when they initiated a new area of inquiry. In total, 189 pages of transcribed notes were produced from the interviews.

Prior to each interview, a questionnaire was completed by participants in order to determine their level of experience with agile and waterfall development. Interviewees indicated their level of experience with 10 systems development techniques associated with waterfall development (e.g. specialised staff roles, working software at the end of a project and importance of technical documentation) and 10 techniques associated with agile development (e.g. pair programming, story cards and collective code ownership). The techniques were not labelled by approach and were randomly ordered in the questionnaire. A six-point scale was used, from 0 (no experience) to 6 (extensive experience). Experience scores ranged from 20 to 58 in agile experience, with an average of 37.5. For waterfall experience, scores ranged from 10 to 60, with an average of 38.7. Based on these results, the consolidated interviewee sample is concluded to have an approximately equal level of experience with both agile and waterfall approaches, minimising concerns regarding an experience bias with one of the ISD approaches.

Data analysis

Defining an analysis strategy is an important step in identifying what data to analyse and why it should be analysed in order to develop theory (Dube & Pare, 2003; Yin, 2009). First, the interviews were reviewed and coded by the first author, with the identified control dimensions used as coding categories (Miles & Huberman, 1994). A sample of seven interviews was reviewed in detail by the second author in order to validate the reliability of coding. All issues were discussed, and coding revisions were made, where necessary. Similar techniques have been used in other studies such as Silva & Hirschheim (2007) as a means of facilitating researcher collaboration to increase data analysis reliability. Because the data collection and analysis stages overlapped, the increased flexibility resulted in the addition of one new control dimension to the corrective control practices category: prioritisation. Miles & Huberman (1994) suggest that this process of revision acts as a refinement of the original conceptual structure of the study and integrates empirically grounded insights generated from the field site. All controls noted within the interview transcripts were coded to one or more of the twelve identified control dimensions.

The final data analysis step was the in-depth examination of the coded data. Cross-case patterns were used to identify within-group similarities and intergroup differences (Dube & Pare, 2003). To achieve this objective, NVivo matrix queries were used to identify instances within the interview transcripts where the characterisation of a systems development approach and a control dimension corresponded to a particular interviewee quote. By comprehensively examining the matrix reports, insights were developed into the nature of control variance across development approaches.

RESULTS

Based on the collected data, two sets of defining themes emerged from our analysis: control objectives vary on the basis of a product or process focus, and control practices vary on the basis of a preventive or detective/corrective focus. These results provide a basis for developing a response to the research question related to why and how control differs across development approaches. In regard to *why* control differs, the data suggest that systems development approaches inherently diverge on the basis of a dominant focus on either the creation of a product or the execution of a process. Depending on the adopted approach, particular control objectives (e.g. quality vs. speed) become increasingly important relative to others. In regard to *how* control differs, our results suggest that development approaches adopt particular control practices that characterise the operational goals and principles of a development project. Where ISD projects are required to avoid deviation from a standard approach, preventive practices are adopted; however, where projects require flexibility and creativity, detective and corrective control practices can be used. This section will outline these themes and provide representative interview quotes that illustrate and validate the assertions.

Control objectives

Our analysis suggests that the first control category, control objectives, varies on the basis of either a product focus or process focus, depending on the development approach. Consistent with Gemino *et al.* (2007), we define a product focus as those control objectives related to the quality of a developed system and its related benefits, while a process focus represents the control objectives related to the efficiency and effectiveness of the development project activities. Therefore, we attributed the functionality, compliance with regulations and quality control objective dimensions as being representative of a product focus and attributed the delivery speed, achievement of budget and risk reduction control objective dimensions to the process focus (see Table 2). The data suggest that where a waterfall development approach is used (i.e. LargeMan), control objectives tend to be dominated by a product rather than a process focus. Alternatively, where an agile development approach is used (i.e. OntCode), control objectives tend to focus on process objectives rather than product objectives. In the case of a hybrid development approach (i.e. InfoOrg), there is evidence of a more balanced view of both product and process objectives.

Control objectives were found to be most closely associated with process-focused control objectives when an agile development approach is used. The following quotes provide an

Table 2. Control objective categorisations

Product-focused control objectives	Process-focused control objectives
Functionality	Speed
Regulatory compliance	Budget
Product quality	Risk reduction

example of interviewee comments that support this assertion. In particular, the comments highlight the importance of task and time management, issue resolution and attainment of efficiency.

I run the daily stand-up meetings. They are very short; they always focus on what was done yesterday, what's going to be done today. And I keep the backlog of the issues and tasks up to date, so I daily report on how much time we spent, what's left – remaining hours or remaining time, as well as how it compares to the estimate from the initial[ization] of the project. (Project Manager, InfoOrg)

In an agile project, because of the short durations, you tend to break your tasks down into fairly small increments. So . . . it's easier for the project manager and for the person responsible to know, 'yeah, I should be finished [with] this by this afternoon and I should work on this and the next four hours I should be working on this'. (Developer, InfoOrg)

A lot of the time you think, 'okay I know how to do this' . . . and then, 'oh, wait a minute you know what, there's actually a better way of doing it' and when you actually hear other people's opinions, then you kind of grow . . . your knowledge base of how to provide a solution to a particular problem. (Developer, OntCode)

People like the process. The daily interaction [and] the daily stand-up meetings are very beneficial to everyone, not just the Scrum Master or the Project Manager. But, there is a sense of collaboration, team work, being on top of all the issues, knowing exactly what's ready, what's not ready. So, we can prioritize quickly and effectively. (Project Manager, InfoOrg)

In contrast, control objectives were found to be most closely associated with product-focused control objectives when a waterfall development approach is used. The following quotes provide an example of interviewee comments that support this assertion. In particular, the comments highlight the importance of delivering the system functionality required by users at a high standard of quality.

[Design documents] clarify exactly what's needed and show the user what they will see. It's kind of like a contract right, you'd say, 'you write this down, this is what you'll see, this is what we're changing'. (Developer, LargeMan)

I think any QA [quality assurance] person will tell you that the better the requirements, the better the job they can do because you know what you should be testing against what. You know . . . the more defined the requirement, the better my job is. (Developer, LargeMan)

So any business functionality that goes in, we schedule a walk through . . . whether it be International or Europe or U.S., schedule a walk through at that date to go through the functionality. So it forces us to make sure that everything is tested and is all working properly before we get to that point. (Developer, LargeMan)

Control practices

The second theme emerging from our analysis indicates that the second control category, control practices, varies on the basis of either preventive control practices or detective/corrective control practices depending on the development approach. Consistent with Cushing (1974) and Srivastava (1985), we define preventive controls as those intended to avoid errors or issues from occurring, while detective and corrective controls are intended to report the presence of an issue and then remedy it. Therefore, we attributed the enforcement, process standardisation and process prioritisation control practice dimensions as being representative of a preventive focus and attributed the active monitoring, personnel interaction and remediation control practice dimensions to the detective/corrective focus (see Table 3). The data suggest that where a waterfall development approach is used (i.e. LargeMan), control practices tend to be primarily preventive in nature, while projects using an agile development approach (i.e. OntCode) tend to rely more heavily on detection and correction control practices. Similar to the findings associated with control objectives, where a hybrid approach is used, control practices exemplify a balance between preventive and detective/corrective practices.

In contrast, control practices were found to be most closely associated with a preventive control focus when a waterfall development approach is used. The following quotes provide an example of interviewees' comments that support this assertion. In particular, the comments highlight the importance of proactive issue avoidance, pragmatism and formal procedures.

We know when we have to do the installation and then from there we will build a time line around that; around scoping, requirements gathering, development and a testing phase. (Business Analyst, LargeMan)

[Management] can look at a project and make the decision if they're going throw good money after bad. If there's something that is floundering, they've built these structures in that can look at it and say 'you need more work on this' or 'no, you can't even go any further'. (Developer, LargeMan)

We use a lot of red/yellow/green to show when stuff is on track or not on track. Some people really take it to heart if things go yellow or go red. (Director, LargeMan)

[The development process is] all set up with a series of gates and so, to pass from conception into feasibility to some of these other broad chronological phases, you have to pass through these gates and you have to have people championing and give presentations and it moves on and people sign-off and it moves to another gate. (Developer, LargeMan)

Table 3. Control practice categorisations

Preventive control practices	Detective/corrective control practices
Enforcement	Active monitoring
Standardisation	Personnel interactions
Prioritisation	Remediation

Control practices were found to be most closely associated with detective and corrective control practices when an agile approach is used. The following quotes provide an example of interviewee's comments that support this assertion. In particular, the comments highlight the importance of collaboration, rapid response and problem resolution.

With agile, it's quick; you are always on top of everything that's happening. You can reprioritize quickly. You can change the scope because you gained more understanding and you share it and then your tester is right with you. (Project Manager, InfoOrg)

Things get done faster because we have a lot of face-to-face communication. We're in the same room; we don't send emails back and forth where they end up in some folder. I'm going to get up and I'm going to walk over to you . . . and ask: 'so what's going on with this, I'm running into this problem, can you come over here and give me a hand?' (Developer, OntCode)

So, right away if you start something and the client says, 'oh wait a minute, I want to do that instead' or 'I don't really like the way it looks', right away you can fix it'. (Developer, OntCode)

Because the daily stand-ups are meant to be very quick and it's more like, 'okay, we came to a halt, we cannot move forward'; so we bring that up . . . then we identify the issue and say, 'okay, you developer and you architect, talk together later today and get that resolved'. (Project Manager, InfoOrg)

ISD CONTROL: A NEW TYPOLOGY

Based on the results discussed above, we propose a new typology of ISD control (see Figure 1). Doty & Glick (1994) define a typology as a conceptually derived, interrelated set of ideal types, which represent configurations of multiple constructs. We propose four ideal

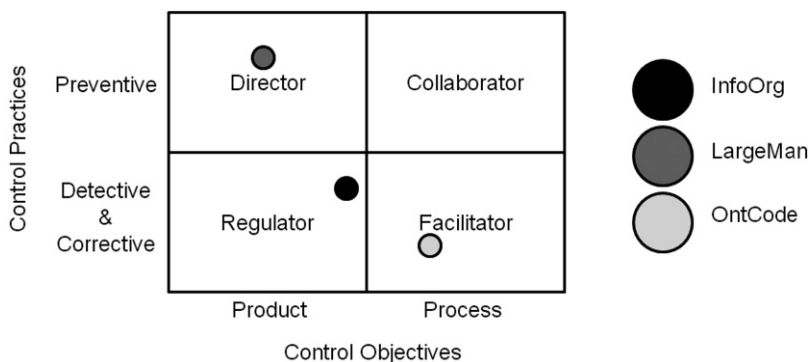


Figure 1. Typology of information systems development control.

types, defined on the basis of differing configurations related to their approaches to control objectives and practices. For each type, we provide a description of its defining characteristics, based on the observations and discussions conducted within the case organisations. Two ideal types are defined for projects that focus primarily on product-related control objectives. These projects are more likely to have an externally focused IT group, either through the provision of products or services to customers outside the organisation or to externally facing departments within the organisation. Where these projects focus primarily on preventive control practices, it represents a 'Director' control type. This type of project is likely to be more risk averse and seeks to more carefully structure controls so as to prevent any control deviations from occurring. The use of extensive technical standards and mandatory operational policies are common within this control type. The 'Director' is best suited to initiatives requiring precise, high-quality systems deliverables based on pre-defined specifications but does not allow for flexibility or a high degree of uncertainty. Systems being developed for use within highly regulated industries (e.g. financial services) or industries requiring a high degree of product accuracy (e.g. aerospace and healthcare) would be consistent with a 'Director' control type. Alternatively, where these projects focus primarily on detective and corrective control practices, it represents a 'Regulator' control type. This type of project tends to be more accepting of risk, where it is likely to lead to increased innovation and technical advancements. Projects of this type will expect control deviations to occur but will plan to correct them on an as-needed, iterative basis. By acting as a hub of information relating to the development product, 'Regulators' allow managers and developers to monitor, interact and remediate issues. Projects benefiting from this control type will have a high degree of uncertainty and change but tend to be unsuited for risk averse initiatives requiring achievement of specific speed and budget constraints. Systems being developed within industries characterised by a rapid rate of change (e.g. consumer products) or by organisations intent on maintaining a competitive advantage provided by an innovative application of technology would be consistent with a 'Regulator' control type.

As well, two ideal types are also defined for projects that focus primarily on process-related control objectives. These projects are more likely to be internally facing, where efficiency and effective project management are of particular importance. Where these initiatives focus primarily on prevention, it represents a 'Collaborator' control type. These projects seek to achieve seamless development activities by proactively and collectively avoiding the occurrence of control issues through process streamlining. 'Collaborators' will pay close attention to budget status and anticipated delivery dates, potentially leading to the removal of product features where process objectives are in jeopardy. Initiatives using this control type are not concerned with the delivery of a perfect system to users; instead, they focus on minimising scope creep and other project risks through rigorous operational management. Large, multi-phased systems initiatives that are of strategic importance to an organisation (e.g. supply chain) are consistent with 'Collaborators', where timely implementation is critical and systems requirements are clear. Alternatively, where these projects focus primarily on detective and corrective control, it represents a 'Facilitator' type. Facilitator projects aim to correct control issues as they occur as a means to allow flexibility for the development team to remain on

schedule and within budget. This control type relies heavily on communication and interaction between team members as well as experienced leadership. 'Facilitator' control types tend to be most appropriate for ISD projects with a high degree of uncertainty, where on time and on budget delivery is required. Where development projects are oriented towards systems used for operational support (e.g. inventory management and procurement) and where requirements are ambiguous, 'Facilitators' can contribute timely, cost-conscious deliverables while maintaining a degree of flexibility.

Based on the results from this study, we would expect that those organisations that have primarily product-related control objectives and that prefer primarily preventive control practices will have the greatest degree of fit with a systems development approach that is highly structured such as waterfall. In contrast, those organisations that have primarily process-focused objectives and utilise detective and corrective practices will have the greatest degree of fit with more flexible development approaches such as agile. Finally, in cases where an organisation is categorised as either a Regulator (i.e. product objectives and detective/corrective practices) or a Collaborator (i.e. process objectives and preventive practices), a hybrid development approach may be most effectively used. However, the four ideal types presented in the proposed typology represent examples of organisations with extreme, 'ideal' control objective and control practice preferences. In practice, organisations may adopt more moderate inclinations towards both control and development approaches.

On that basis, the proposed typology presented in Figure 1 includes three highlighted points corresponding to each of the three organisations participating in this study. LargeMan is considered to be a 'Director' type, InfoOrg is considered to be a 'Regulator' and OntCode is viewed as a 'Facilitator'. Our findings suggest that as increasingly 'pure' forms of a development approach are used, such as with LargeMan (i.e. waterfall) and OntCode (i.e. agile), the strength of the distinctions made between categories of control objectives and practices are progressively clearer. However, in the case of organisations using a hybrid approach, such as InfoOrg, the evidence corresponding to control dimensions within disparate themes are more evident. This balanced view of product vs. process objectives and preventive vs. detective/corrective practices increasingly allows the pursuit of a wider range of control objectives and a wider selection of control practices than appears to be typical in a primarily agile or waterfall approach.

DISCUSSION

Control typology application

The proposed typology can help to demonstrate the important differences in control that exist within ISD projects as well as within other organisational initiatives. By understanding what type of control structures are used within an IT department, practitioners can use this typology as a tool to consider the nature of control that is used as well as the degree of alignment between the controls they use on systems development initiatives and the

controls used elsewhere in the organisation. For example, where IT initiatives tend to use the Regulator type, while business initiatives regularly use the Collaborator type, it may be evidence of a misalignment of control portfolios. Past research suggests that where controls are inconsistently applied, it can cause confusion and frustration for employees (Orlikowski, 1991). There is the potential for performance impacts from such a situation, though further research is required to investigate this important phenomenon. By providing a new perspective on the control characteristics of systems development, practitioners may wish to reconsider the reasons why they choose particular control techniques for use on their projects. Where a control objective or control practice can be selected that is more consistent with other control choices for a project or when it is more consistent with other control choices made within the organisation, this typology can encourage practitioners to make more effective control choices.

Going beyond modes

This typology also contributes to advancing research related to IS control by providing a perspective on IS control that differs from the standard view in the field related to control modes, as has been called for by past research (Kirsch, 2004). Though we recognise that control modes have contributed extensively to IS control research, there is evidence that when control modes are used to assess new forms of systems development, results have been inconsistent. For example, two articles recently published in the Information Systems Research special issue on agile methods reveal seemingly contradictory findings related to the control of agile projects. Maruping *et al.* (2009) conclude that outcome control (i.e. monitoring the output of employees) is critical in agile environments to enable a focus on project objectives in light of changing requirements; however, Harris *et al.* (2009) argue that outcome control provides an insufficient description of agile environments and proposes the concept of emergent outcome control as a new concept to explain agile environments.

One plausible explanation for such conflicting results in the literature is that control modes do not sufficiently differentiate between the control objectives and control practices inherent in all systems development approaches. However, we argue that our typology can help to reconcile past conflicting results and extend the dialogue on control in systems development. For example, Maruping *et al.* (2009) focus exclusively on the control objective of using agile practices to improve software quality. The achievement of quality is considered to be a product-focused control objective in our typology (i.e. Director or Regulator types). In comparison, Harris *et al.* (2009) focus on emergent outcome controls consisting of two mechanisms. The first mechanism is scope boundaries, a risk management mechanism, which is considered a process control objective in our typology (i.e. Collaborator or Facilitator types). The second mechanism is ongoing feedback, which is considered a personnel interaction control and is associated with detective/preventive control practices (i.e. Regulator or Facilitator types). As a result of the differing control objectives and control practices investigated in these two studies, we suggest that it is unsurprising they arrive at different conclusions regarding the relative importance of a single control mode. Although each paper explores the

use of agile development approaches in organisations, we argue that our typology uncovers that they are focused on projects that fundamentally differ on the basis of control objectives and control practices. This may help to explain the contradictory findings and highlight the insights that our typology can provide.

By considering alternative dimensions of control, as we do in this study, researchers can increasingly refine the core principles that drive systems development initiatives. As these principles evolve over time, corresponding with advancing ISD trends, we argue that new perspectives on control can help researchers better understand how and why particular approaches are adopted. The typology presented above helps to move beyond the concept of control modes by considering the control objectives and practices that correspond to different development approaches.

CONCLUSIONS

The objective of this paper was to conduct an in-depth exploration of the dimensions of systems development control as a means to derive insights related to how development approaches control ISD projects differently. Two important contributions stem from this research. First, our findings suggest that when control is viewed using the categories and dimensions proposed in this study, unique insights can be generated regarding the systems development process that go beyond the current control modes approach. By drawing on the organisational control literature, we find support in our assertions that a new typology consisting of control objectives and control practices can help to reconcile past contradictory findings related to systems development control. Previous research examining the relationship between development approach elements such as *livari et al.* (2000), has been characterised as taxonomic theory (Gregor, 2006); however, this study builds on these classifications by incorporating the construct of control. By empirically establishing a typology of ISD control, the findings from this study represent a novel contribution to control theory in systems development.

Second, our proposed typology provides a basis for practitioners to more thoroughly evaluate the nature of control used within their systems development projects. By constructing an accessible and understandable typology of systems development control in terms that are relevant to practice, this study can help to inform IS managers on the alternatives and potential implications of particular control choices. By enabling practitioners to categorise the type of control they utilise on their own projects, the typology represents a useful tool that can enable a straightforward approach to determining the degree of alignment between the types of control used across an organisation.

This study has two primary limitations. First, our findings are based on data collected at only three organisations. Though we attempt to enhance generalisability by selecting a diverse set of participants and conducting cross-case analysis, organisations that are notably distinct from our sample may uncover alternative interpretations of our typology. For example, the participants in our study using an agile approach tended to use detective/corrective control practices

as a means to achieve process objectives. Other organisations utilising markedly different development approaches may find an alternative control objectives and practices pairing. Second, we focus exclusively on two ISD approaches, waterfall and agile. Although we believe that other development approaches are likely to exhibit distinct collections of control objectives and practices, additional study is required to determine if supplementary typology dimensions are required to capture any unique control elements.

Future research can build on the findings noted in this study. First, subsequent studies could further validate and expand on the typology of ISD control proposed in this paper. Our model contributes to building a better understanding of the interrelationship between control and the systems development process; however, other control categories and dimensions may exist that can provide further insights. Additional examination of the four ideal control types within a wider collection of organisations could validate and enhance the proposed typology. As well, our results may highlight an opportunity to investigate the extent that organisations use controls adopting similar objectives and practices across multiple ISD projects or other IS processes. Such research could identify the benefits or drawbacks of using controls with uniform characteristics not just within a single project but also on a broader scale. Finally, we address the issue of ISD control but do not directly examine the complex role of power, politics and authority that can influence how and why systems are developed within organisations. Future studies could consider the role of ISD control as it applies from the board room to the server room as a means to better understand the complexities that can influence the process of systems development.

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